SUPPLYNEXUS AI

AI Intelligence for the Connected Supply Chain

Phase 0 — Product Discovery

Version 0.1 · Discovery Draft
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Product owner: Aman Agrawal
Status: **CONTINUE PRODUCT DISCOVERY**
Next phase: Phase 1 — Requirements & PRD (not started)

Contents

1. Document Purpose
2. Product Identity
3. Product Vision
4. Problem Discovery
5. Problem Statement
6. Target Organizations
7. Target Users
8. Personas
9. Buyer / Stakeholder Map
10. Supply Chain Pain Points
11. Current Workflow Analysis
12. Existing Solutions
13. AI Opportunities
14. Non-AI Opportunities
15. Core Use Cases
16. MVP Discovery
17. Initial MVP Boundary
18. Product Scope
19. Non-Goals
20. Product Differentiation Hypotheses
21. Business Value Hypotheses
22. Success Metrics
23. Assumptions Register
24. Validation Strategy
25. Major Risks
26. Product Principles
27. Future Product Vision
28. 30-Second Product Explanation
29. 2-Minute Product Explanation
30. 5-Minute Product Explanation
31. Phase 0 Decision Summary
32. Open Questions
33. Phase 1 Inputs

Phase 0 Final Quality Check

Phase 0 Status

Phase 0 Handoff

Field	Value
Document	Phase 0 — Product Discovery
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SDLC stage	Stage 1 — Product
Phase status	**CONTINUE PRODUCT DISCOVERY** (see Section 31 and the Handoff)
Next phase	Phase 1 — Requirements & PRD (not started)

How to read this document — evidence labels

Every important statement carries a label so that hypotheses are never mistaken for research.

Label	Meaning
[SOURCE-DERIVED FACT]	Stated directly in the SupplyNexus AI project material (project brief, Master SDLC file, Phase 0 prompts). These are *facts about the project's intent*, not facts about the market.
[INFERENCE]	A reasoned conclusion drawn from the source material.
[ARCHITECTURAL RECOMMENDATION]	A product or architecture recommendation made in this document. The product owner may accept or reject it.
[GENERAL ENGINEERING KNOWLEDGE]	Widely-held industry or engineering knowledge, not specific to SupplyNexus.
[ASSUMPTION]	Believed to be true but not evidenced.
[TO VALIDATE]	Must be confirmed with users, practitioners, experiments or data before it can be relied on.
[UNKNOWN]	Not established by any source.

> **Important disclosure.** No customer interviews, user research, market statistics or competitive analysis have been performed or supplied. Nothing in this document should be read as customer evidence. All market, user, buyer and value statements are hypotheses.

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1. Document Purpose

This document establishes **why** SupplyNexus AI should exist, **who** it is for, **what** problem it addresses, **what value** it could provide, **what** the initial product should and should not do, and **which assumptions** still require validation. [SOURCE-DERIVED FACT]

It is deliberately **not**:

- a PRD (Phase 1),
- a domain or SaaS business model (Phase 2),
- a system, data, AI, security or API architecture (Phases 3–7),
- an implementation, coding or deployment plan (Phase 8 onwards).

Technologies listed in the project brief (Angular, .NET, Python/FastAPI, PostgreSQL/pgvector, Amazon Bedrock, MCP, AgentCore, Kong, Redis, S3, Opik, OpenTelemetry, Docker, Terraform, GitHub Actions, AWS) are a **potential** direction and are **not** decided here. [SOURCE-DERIVED FACT] — the source explicitly states that "none of these technologies are automatically mandatory."

The document also serves a second, source-stated purpose: SupplyNexus AI is both a real SaaS product *and* the owner's AI-engineering learning platform. [SOURCE-DERIVED FACT] Where these two goals pull in different directions, this document says so explicitly rather than hiding the tension.

2. Product Identity

Attribute	Value	Status
Product name	SupplyNexus AI	SOURCE-DERIVED FACT
Tagline	"AI Intelligence for the Connected Supply Chain"	SOURCE-DERIVED FACT
Category	Supply Chain Intelligence SaaS (multi-tenant)	SOURCE-DERIVED FACT
Delivery model	Multi-tenant SaaS	SOURCE-DERIVED FACT
Interaction model	Natural-language questions that are answered through the most appropriate governed means (knowledge, data, or investigation)	SOURCE-DERIVED FACT
Product nature	Decision-support / investigation tool — **not** a system of execution	INFERENCE
Dual purpose	Real SaaS product first, AI-engineering learning platform second	SOURCE-DERIVED FACT
Builder	Individual contributor	SOURCE-DERIVED FACT

Working product thesis (from source)

> *SupplyNexus AI is an enterprise supply-chain intelligence platform designed to help supply-chain teams understand, query and investigate fragmented operational data and knowledge through a governed AI interface that produces evidence-backed business insights.* [SOURCE-DERIVED FACT]

3. Product Vision

3.1 What SupplyNexus AI is

A governed, natural-language "intelligence layer" that sits **alongside** existing supply-chain systems (WMS, ERP, OMS, TMS, inventory, supplier and transportation systems, databases, APIs, spreadsheets, documents, object storage) and helps people **understand, query and investigate** what is happening in their operation. [SOURCE-DERIVED FACT]

3.2 The long-term problem it addresses

Supply-chain information is distributed across many systems and document stores. Answering an operational question — especially a *why* question — often requires a person to know which systems to look in, extract data, reconcile it, and apply domain knowledge. [INFERENCE from source's emphasis on "fragmented operational data and knowledge"]

3.3 What users should ultimately be able to accomplish

- Ask a factual question about their operation and receive a correct, traceable answer. [SOURCE-DERIVED FACT]
- Ask what a term, policy or procedure means *in their organization* and receive an answer grounded in their own documents. [INFERENCE]
- Ask *why* something happened (e.g. "Why was SKU-10023 oversold?") and receive an explanation supported by evidence the user can inspect. [SOURCE-DERIVED FACT]
- Eventually, receive recommendations — and, much later, controlled and approved actions. [SOURCE-DERIVED FACT for recommendations; controlled actions are future vision]

3.4 Why this is different from "building a chatbot"

A generic chatbot	SupplyNexus AI (intended)
Answers from general model knowledge	Answers from the **tenant's** governed data and documents [SOURCE-DERIVED FACT]
Output is plausible text	Output is an answer **plus evidence** (records, documents, calculations) [SOURCE-DERIVED FACT]
One mechanism for every question	Distinguishes deterministic data retrieval, knowledge retrieval and investigation [SOURCE-DERIVED FACT]
Often unrestricted access to whatever it is connected to	Governed, read-only-first, tenant-scoped access; no arbitrary queries [SOURCE-DERIVED FACT]
Success = "sounds right"	Success = measurably correct, grounded, traceable [SOURCE-DERIVED FACT]

3.5 Why evidence and governed access matter (product view)

In operations, a wrong number or wrong root cause leads to wrong actions (wrong reorder, wrong escalation, wrong supplier blamed). Therefore **trust is a product feature**, not an engineering nicety: users must be able to see *where* an answer came from, and organizations must be confident that the product cannot see or leak data it should not. [INFERENCE] [GENERAL ENGINEERING KNOWLEDGE]

3.6 Conceptual value flow (product view — not an architecture)

```
User question
   ↓
SupplyNexus AI  (understands what kind of question it is)
   ↓
AI reasoning    (only where reasoning is actually needed)
   ↓
Governed capabilities  (approved, permission-checked ways to get information)
   ↓
Supply-chain data / knowledge  (the customer's own systems and documents)
   ↓
Evidence        (the records, documents and calculations behind the answer)
   ↓
Business insight
   ↓
Recommended action  (a human decides)
```

3.7 How the product could evolve

Understand (knowledge) → **Query** (data) → **Investigate** (why) → **Recommend** → **Anticipate** (risk/anomaly) → **Act, with approval** (controlled actions). [INFERENCE from source future capabilities] Only the first three are candidates for MVP (Section 16); the rest are future vision (Section 27).

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4. Problem Discovery

4.1 Problem, symptoms, root causes, consequences

Layer	Description	Status
Core problem	Supply-chain teams struggle to get **timely, trustworthy answers** — especially explanations of *why* something happened — because the relevant data and knowledge are fragmented across systems, documents and people.	INFERENCE (supported by source thesis)
Symptoms	Repeated ad-hoc data requests; spreadsheet exports; "who knows how to check this?" questions; slow exception follow-up; different teams quoting different numbers for the same metric.	ASSUMPTION / TO VALIDATE

Layer	Description	Status
Root causes	(1) Data lives in multiple systems with different identifiers and timing; (2) domain knowledge lives in documents and in people's heads; (3) querying data requires SQL/BI skills or IT tickets; (4) cross-system investigation is manual; (5) fixed reports answer *known* questions, not ad-hoc ones.	INFERENCE + GENERAL ENGINEERING KNOWLEDGE
Consequences	Slower decisions; slower exception resolution (stockouts, overselling, backlog, delays); analyst time spent gathering rather than analysing; dependency on a few experts; inconsistent explanations to management and customers.	ASSUMPTION / TO VALIDATE

4.2 Why these questions are hard to answer

1. **Systems are specialised.** WMS knows the warehouse, OMS knows orders, ERP knows purchasing/finance, TMS knows transport. A *why* question usually crosses at least two of them. [GENERAL ENGINEERING KNOWLEDGE]
2. **Identifiers and timing differ.** An order, shipment, SKU and warehouse may be keyed differently in each system, and each system updates on its own schedule. [GENERAL ENGINEERING KNOWLEDGE]
3. **Knowledge is unstructured.** Policies, SOPs, supplier agreements and definitions (e.g. how *this* company computes safety stock) live in documents. [INFERENCE]
4. **Skill barrier.** The person who has the question is often not the person who can write the query. [ASSUMPTION / TO VALIDATE]
5. **Investigation is a sequence, not a query.** Root-cause analysis requires forming hypotheses, pulling several pieces of evidence and correlating them. [INFERENCE]

4.3 What happens when information is distributed

- The *first* cost is **time to find** (which system? which report? who owns it?).
- The *second* is **time to reconcile** (numbers disagree; timestamps differ).
- The *third* is **risk of wrong conclusion** (the analysis is only as good as the data the analyst happened to find). [INFERENCE] — magnitudes are **[UNKNOWN / TO VALIDATE]**.

4.4 Skeptical challenge — is the problem painful enough?

Problem	The source asserts the problem (fragmentation) but provides no evidence of *pain magnitude* or *willingness to pay*. The discovery canvas itself marks "Actual customer pain magnitude" as **UNKNOWN**.
Impact	If pain is moderate, organizations will continue with BI dashboards, Excel and analysts; SupplyNexus becomes "nice to have."
Example	"How many orders were returned last month?" is already on most returns dashboards; automating it with AI adds little.
Recommendation	Anchor the product on **ad-hoc *why* questions and exceptions** (where fixed reports fail), not on questions that dashboards already answer. Validate pain with time-on-task evidence (Section 24).
Alternative	Position primarily as a self-service data-query tool for non-technical users.
Trade-off	Self-service querying is easier to build but more crowded and less differentiated; investigation is more valuable but harder, riskier and slower to build.

5. Problem Statement

Field	Statement
Problem	Supply-chain and operations teams cannot quickly obtain trustworthy answers and explanations about their operation because the needed data and knowledge are spread across multiple systems, reports, documents and people. Explaining *why* an exception occurred (oversell, stockout, backlog, delay, discrepancy) is especially slow and manual.
Who experiences it	Primarily supply-chain / operations analysts and the managers who depend on their answers. **[TO VALIDATE]**

Field	Statement
When it occurs	When an ad-hoc question is not covered by an existing report; when an exception needs a root cause; when a definition or policy must be checked; when management needs an explanation quickly. **[INFERENCE / TO VALIDATE]**
Current consequence	Manual, multi-step investigation (find system → export → combine in Excel/SQL → ask other teams → reconcile → explain). **[CURRENT WORKFLOW HYPOTHESIS — Section 11]**
Business impact	Delayed decisions and exception resolution, analyst capacity consumed by data gathering, dependency on a few experts, inconsistent explanations. **[ASSUMPTION — magnitude UNKNOWN]**
Evidence from source	The product thesis ("understand, query and investigate fragmented operational data and knowledge"); example questions ("Why was SKU-10023 oversold?", "Which warehouse has the highest backlog?"); the discovery canvas ("Fragmented supply-chain information makes investigation difficult" — *hypothesis supported by source*).
Confidence	**Medium** that the problem exists in general (widely recognised in the industry — GENERAL ENGINEERING KNOWLEDGE); **Low** on who feels it most, how often, and how much they would pay.
Validation required	Who has the problem most acutely; frequency of ad-hoc/why questions; time per investigation; existing tools used; whether evidence-backed answers would change trust and adoption; who pays.

6. Target Organizations

All entries below are a **TARGET MARKET HYPOTHESIS**. None is a confirmed market.

Organization type	Why SupplyNexus might fit	Data / integration complexity	Status
Mid-size to large distributors	Many SKUs, multiple warehouses, suppliers and carriers; frequent inventory and fulfilment exceptions	Medium–High (ERP + WMS + TMS typical)	TARGET MARKET HYPOTHESIS
Omnichannel retailers / e-commerce operators	Overselling, returns, backlog and delivery delays are visible and customer-facing	Medium–High (OMS + WMS + marketplaces)	TARGET MARKET HYPOTHESIS
3PLs (third-party logistics)	Operate warehouses for many clients; must explain performance and exceptions to customers	High (multi-client WMS, SLA reporting)	TARGET MARKET HYPOTHESIS
Manufacturers	Supplier performance, procurement delays, component shortages	High (ERP/MRP heavy)	TARGET MARKET HYPOTHESIS
Logistics / transportation organizations	Shipment delays, carrier performance	Medium (TMS-centric)	TARGET MARKET HYPOTHESIS
Warehouse operators	Backlog, throughput, inventory accuracy	Medium (WMS-centric)	TARGET MARKET HYPOTHESIS

6.1 Characteristics that likely make SupplyNexus valuable [INFERENCE / TO VALIDATE]

- **Three or more** operational systems that must be combined to answer common questions.
- **High exception volume** (oversells, stockouts, late shipments, discrepancies).
- A **dedicated analyst/ops team** fielding ad-hoc questions from managers.
- **Documented** policies/SOPs that people frequently need to consult.
- **Moderate data maturity**: data is accessible (database or API) but not fully unified.
- **Openness to governed AI** (security review is possible, not a blanket ban).

6.2 Characteristics that likely make SupplyNexus *less* valuable [INFERENCE]

- A single all-in-one system with strong built-in reporting.
- A mature, unified data warehouse with a strong BI/self-service culture.
- Very low data maturity (data not accessible; the problem is data quality, not access).
- Regulated environments that forbid external AI processing of operational data **[TO VALIDATE per organization]**.

6.3 Skeptical challenge — target organization is too broad

Problem	Six organization types with different systems and problems cannot all be served by one MVP.
Impact	Unfocused use cases, generic data model, weak product-market fit signal.
Example	A 3PL's key question ("which client's SLA is at risk?") differs from a retailer's ("why was this SKU oversold?").
Recommendation	Choose **one beachhead** for the MVP. The source examples (orders, returns, SKU overselling, warehouse backlog) most naturally fit an **omnichannel retailer / e-commerce operator or distributor running its own warehouses**. [ARCHITECTURAL RECOMMENDATION — TO VALIDATE]
Alternative	3PL beachhead (multi-client explanation needs are strong).
Trade-off	The retail/distribution beachhead aligns with available public data (e.g. Olist is e-commerce order data — SOURCE-DERIVED FACT that Olist is a candidate dataset) but may be more crowded; the 3PL beachhead may have sharper pain but adds multi-client complexity on top of multi-tenancy.

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7. Target Users

User	Classification	Rationale	Status
Supply Chain Analyst	**Primary User**	Fields ad-hoc questions, runs investigations, closest to the problem	TO VALIDATE (source canvas: "Initial user: supply-chain / operations analyst — TO VALIDATE")
Operations Analyst	**Primary User**	Same job-to-be-done in many organizations	TO VALIDATE
Inventory Manager	**Secondary User**	Owns inventory exceptions (oversell, stockouts, discrepancies)	TO VALIDATE
Warehouse Manager	**Secondary User**	Backlog, throughput, warehouse exceptions	TO VALIDATE
Supply Chain Manager / Operations Manager	**Secondary User**	Consumes answers; asks "why"; makes decisions	TO VALIDATE
Logistics Manager	**Occasional User**	Shipment delays; transport questions	TO VALIDATE
Procurement Manager	**Occasional User**	Supplier performance; PO delays	TO VALIDATE
Executive / Director	**Occasional User** (and potential buyer)	Needs summaries and explanations, not data work	TO VALIDATE
Data Analyst / Business Analyst	**Technical User**	May validate answers, curate definitions, compare with BI	TO VALIDATE
IT / Data Team	**Administrative User**	Connects sources, manages access, approves security	TO VALIDATE
Tenant Administrator	**Administrative User**	Manages users, roles, documents	INFERENCE (implied by multi-tenant SaaS)
Customer-service / account teams at 3PLs	**Potential Future User**	Explain exceptions to external clients	ASSUMPTION

Recommendation: treat the **Supply Chain / Operations Analyst** as the single primary MVP persona, and the **Operations / Supply Chain Manager** as the primary *consumer* of the analyst's output. [ARCHITECTURAL RECOMMENDATION — TO VALIDATE]

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8. Personas

> All personas are **hypothetical constructs** built from source examples and general industry knowledge. They are not derived from interviews. **[ASSUMPTION / TO VALIDATE]**

8.1 Persona A — "Priya", Supply Chain / Operations Analyst (Primary)

Attribute	Description
Responsibilities	Monitor operational KPIs; answer ad-hoc questions from managers; investigate exceptions; prepare explanations and reports
Goals	Answer quickly and correctly; spend time on analysis instead of data gathering; be trusted
Pain points	Data spread across systems; repetitive extracts; reconciling conflicting numbers; slow access to other teams' data; urgent "why" requests
Current workflow	Dashboards → exports → Excel / SQL → emails/chats with other teams → explanation (Section 11)
Information needs	Orders, inventory positions and movements, warehouse status, shipments, returns, supplier performance, definitions/policies
Typical questions	"How many orders were returned last week?", "Which warehouse has the highest backlog?", "Why was SKU-10023 oversold?", "What is our safety-stock policy for fast movers?"
Decisions made / supported	Escalations, root-cause statements, recommendations to managers
Potential SupplyNexus value	Faster answers, fewer manual extracts, evidence packs she can forward, consistent definitions

8.2 Persona B — "Daniel", Operations / Supply Chain Manager (Secondary; primary consumer)

Attribute	Description
Responsibilities	Service levels, cost, backlog, escalations, team performance
Goals	Know what is wrong, why, and what to do — quickly
Pain points	Waiting for analysts; explanations that differ by source; unclear root causes
Current workflow	Reviews dashboards; asks analysts; joins exception calls
Information needs	Summaries, trends, root causes, risk of repeat
Typical questions	"Why are we late on shipments from WH-001?", "Is the backlog getting worse?"
Decisions	Resource allocation, escalations, supplier/carrier conversations, policy changes
Potential value	Direct self-service answers for simple questions; trustworthy explanations for complex ones

8.3 Persona C — "Marcus", Inventory Manager (Secondary)

Attribute	Description
Responsibilities	Stock accuracy, replenishment policy, overselling/stockout prevention
Pain points	Discrepancies between systems; timing issues between order capture and stock updates
Typical questions	"Why does WMS show 40 units but OMS allowed 55 orders?", "Which SKUs were oversold this week?"
Potential value	Faster discrepancy investigation with timeline evidence

8.4 Persona D — "Elena", Warehouse Manager (Secondary)

Attribute	Description
Responsibilities	Throughput, backlog, labour, dock scheduling
Typical questions	"Which warehouse has the highest backlog?", "What's driving today's backlog at WH-002?"
Potential value	Backlog drivers and comparisons without waiting for reports

8.5 Persona E — "Sam", IT / Data Owner (Administrative / Technical)

Attribute	Description
Responsibilities	Data access, integrations, security approval, cost

Attribute	Description
Concerns	Data leakage, uncontrolled queries against production systems, prompt injection, auditability, AI cost
Questions they ask of the product	"What exactly can it access? Is it read-only? Where does data go? Can I audit it? Can one tenant see another's data?"
Potential value	A governed alternative to users pasting data into unapproved public AI tools **[ASSUMPTION / TO VALIDATE]**

8.6 Persona F — Executive / Director (Occasional; potential economic buyer)

Attribute	Description
Interests	Service level, cost, working capital, customer satisfaction
Use	Summaries and explanations; rarely hands-on
Value	Faster, more consistent operational explanations; analyst productivity

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9. Buyer / Stakeholder Map

> No organizational structures are known. Every role below is **[TO VALIDATE]**; the economic buyer is **[UNKNOWN]**.

Role	Likely holder (hypothesis)	Interest	Concern	Expected value	Potential objection
Economic Buyer	VP / Director of Supply Chain or Operations — **UNKNOWN**	Productivity, service levels, cost	ROI, adoption	Faster decisions; analyst capacity	"Our BI tool / ERP vendor already does this."
Business Sponsor	Head of Operations / Supply Chain Manager	Fewer escalations; faster root cause	Disruption to team	Time back; consistent answers	"My team won't trust AI answers."
Primary User	Supply-chain / operations analyst	Less manual work	Accuracy; being blamed for wrong AI answers	Evidence-backed answers	"I still have to verify everything."
Technical Buyer	Head of IT / Enterprise Architecture	Fit with landscape	Integration effort; vendor lock-in	Governed access layer	"Another integration to maintain."
IT Owner	Integration / platform team	Stable, supportable integrations	Load on production systems	Read-only, controlled access	"It must not query production directly."
Data Owner	Data / analytics team	Definition consistency	Conflicting metrics vs BI	Shared definitions	"Numbers must match our warehouse."
Security Owner	CISO / security team	Data protection	Leakage, prompt injection, tenant isolation, third-party AI processing	Auditability, least privilege	"Where is data processed? Is it used to train models?"
AI Governance Owner	Risk / compliance / AI governance (if exists)	Responsible AI	Hallucination, accountability	Evaluation evidence, human-in-the-loop	"How do you measure accuracy?"
Procurement	Procurement department	Terms, price, vendor risk	Vendor viability (a new product built by an individual)	Clear pricing	"Vendor is too early / too small."
Executive Stakeholder	COO / CSCO	Strategic visibility	Cost vs impact	Better operational decisions	"Show measurable outcomes."

9.1 Skeptical challenge — buyer is unknown

Problem	No buyer is identified; the source canvas marks Buyer and Willingness to pay as **UNKNOWN / TO VALIDATE**.
Impact	Without a buyer, value metrics and MVP priorities may optimise for the user's convenience rather than the outcome someone will pay for.
Example	Analysts may love natural-language querying; a VP may only pay for measurable reduction in exception-resolution time.
Recommendation	State an explicit **buyer hypothesis** for Phase 1: *Head of Supply Chain / Operations* at the beachhead organization, buying for **faster exception investigation and analyst productivity**. Mark it TO VALIDATE. [ARCHITECTURAL RECOMMENDATION]
Alternative	IT / data leader as buyer (positioning: "governed AI access layer").
Trade-off	The business buyer values outcomes and domain depth; the IT buyer values governance and integration. Each implies different MVP emphasis.

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10. Supply Chain Pain Points

> Pain points are **hypotheses** derived from source examples and general industry knowledge. Severity and frequency are **[TO VALIDATE]**. "AI suitability" describes whether AI adds value *beyond* deterministic software.

#	Pain point	Who experiences it	Current workaround (hypothesis)	Business consequence	Potential product opportunity	AI suitability	Validation requirement
P1	**Overselling** (orders accepted beyond available stock)	Inventory Mgr, Analyst, Customer Service	Compare OMS orders with WMS/inventory snapshots manually	Cancellations, customer dissatisfaction, marketplace penalties	Evidence-backed oversell investigation (timeline of stock vs orders)	**High** for explanation; **none** for the stock arithmetic	How often; how long an investigation takes today
P2	**Inventory discrepancies** between systems	Inventory Mgr, Analyst	Exports + Excel reconciliation	Wrong availability, wrong replenishment	Discrepancy investigation across sources	**Medium–High** (correlation/explanation)	Which systems disagree most; root-cause patterns
P3	**Stockouts**	Inventory Mgr, Planner	Reports + supplier follow-up	Lost sales	Stockout explanation (demand spike? late PO? wrong policy?)	**Medium**	Is explanation or prediction the real need?
P4	**Warehouse backlog**	Warehouse Mgr, Ops Mgr	WMS dashboards, shift reports	Late shipments, SLA breaches	Backlog comparison + driver explanation	**Low** for ranking (deterministic); **Medium** for *why*	Do dashboards already solve this?
P5	**Shipment delays**	Logistics Mgr, Customer Service	TMS / carrier portals, emails	SLA breaches, customer complaints	Delay investigation across order, warehouse and carrier events	**Medium**	Carrier data availability

#	Pain point	Who experiences it	Current workaround (hypothesis)	Business consequence	Potential product opportunity	AI suitability	Validation requirement
P6	**Returns** volume and reasons	Analyst, Returns Mgr	Returns reports, Excel	Margin loss, quality issues	NL querying + reason analysis	**Low** for counts; **Medium** for reason text analysis	Are return reasons captured as text?
P7	**Supplier performance / late POs**	Procurement Mgr	ERP reports, scorecards in Excel	Shortages, expediting cost	Supplier performance query + explanation	**Low–Medium**	Data availability
P8	**SLA violations**	Ops Mgr, 3PL account teams	Manual SLA reports	Penalties, customer churn	SLA-breach explanation	**Medium**	Beachhead-dependent
P9	**Demand spikes**	Planner, Inventory Mgr	Reports, intuition	Stockouts, overselling	Spike detection & explanation	**Medium** (future: anomaly detection)	Out of MVP (forecasting)
P10	**Operational exceptions** in general	Analyst, Ops Mgr	Manual triage	Slow resolution	Exception investigation workflows	**High** (multi-step reasoning)	Which exception types matter most
P11	**Ad-hoc reporting requests**	Analyst (receives), Managers (ask)	Analyst writes SQL / builds Excel	Analyst capacity consumed	Governed NL data querying	**Medium** (NL→query translation)	Volume of ad-hoc requests
P12	**Data fragmentation**	All	Knowing "who to ask"	Slow, inconsistent answers	Unified governed access across sources	**Low** (integration problem, not an AI problem)	Number and type of systems
P13	**Manual investigation**	Analyst	Multi-step manual process	Time, inconsistency, dependency on experts	Guided / agentic investigation with evidence	**High**	Time-on-task measurement
P14	**Finding policy / SOP / definition knowledge**	Analyst, new staff, managers	Search shared drives; ask colleagues	Inconsistent application of policy	Grounded knowledge assistant over tenant documents	**High** (language, retrieval)	Do tenants have useful documents?
P15	**Procurement / transportation investigation**	Procurement, Logistics	ERP/TMS reports	Cost, delays	Future domain investigations	**Medium**	Post-MVP

Observation. The pain points with the **highest AI suitability** (P1, P10, P13, P14) are *explanation* and *knowledge* problems. Those with the **lowest** (P4 ranking, P6 counts, P12) are *deterministic* or *integration* problems where AI should not be the core. [INFERENCE]

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11. Current Workflow Analysis

> **CURRENT WORKFLOW HYPOTHESIS** — constructed from source examples and general industry knowledge. Not observed. Must be validated through workflow observation (Section 24).

11.1 Hypothesised investigation workflow — "Why was SKU-10023 oversold?"

Step	Activity	Characteristics
1	Receive the question (email/chat/meeting) from a manager or customer-service	Manual

Step	Activity	Characteristics
2	Decide which systems hold relevant data (OMS, WMS, inventory, ERP)	Knowledge-heavy
3	Open reports/dashboards in each system	Manual, cross-system
4	Export order history and stock history for the SKU	Manual, repetitive, data-heavy
5	Combine exports in Excel or run SQL	Slow, error-prone, data-heavy
6	Notice inconsistencies (timing, identifiers, units)	Error-prone
7	Contact another team (warehouse, IT, e-commerce) for missing information	Slow, cross-system
8	Check policies (e.g. safety stock, allocation rules, channel buffers)	Knowledge-heavy
9	Form and test hypotheses (sync delay? receiving error? manual adjustment? demand spike?)	Reasoning-heavy
10	Prepare an explanation with supporting evidence	Manual, repetitive
11	Recommend or take action	Human decision

11.2 Classification of workflow effort

Characteristic	Steps	Product implication
Manual	1, 3, 4, 7, 10	Automation/assistance candidates
Repetitive	4, 5, 10	Deterministic automation first
Slow	5, 7	Governed access to data reduces waiting
Error-prone	5, 6	Deterministic calculations, consistent definitions
Cross-system	2, 3, 7	Integration value (not AI value)
Knowledge-heavy	2, 8	Knowledge retrieval (AI suitable)
Data-heavy	4, 5	Governed data access (AI only for intent translation)
Reasoning-heavy	9	Investigation (AI suitable, evidence required)

11.3 Simple question workflow — "Which warehouse has the highest backlog?"

Question → open WMS dashboard → sort/filter → answer. **This may already be fast.** The product opportunity is only meaningful if the question is **not** on an existing dashboard, spans systems, or is asked by someone without dashboard access/skills. [INFERENCE — TO VALIDATE]

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12. Existing Solutions

> **COMPETITIVE VALIDATION REQUIRED.** No competitive research was provided. The table describes solution *categories* using general industry knowledge. No claims are made about specific vendors or products.

Existing approach	What it already solves well	Where friction may remain	What SupplyNexus could potentially improve
ERP reports	Standard transactional and financial reporting	Rigid; single-system; requires navigation expertise	Cross-system context; NL access
WMS / OMS / TMS dashboards	Operational monitoring within one domain	Siloed; answer "what", rarely "why"	Cross-system *why* explanations
BI tools (dashboards, self-service analytics)	Known KPIs, trends, governed metrics	Ad-hoc questions need skills or new dashboards; weak at multi-step investigation	Ad-hoc NL questions; investigation workflows; knowledge + data in one place

Existing approach	What it already solves well	Where friction may remain	What SupplyNexus could potentially improve
SQL	Precise, flexible	Requires skill and access; risky on production	Governed, validated query access for non-SQL users
Excel	Universal, flexible	Manual, error-prone, not auditable	Reproducible evidence
Data warehouses / lakes	Integrated, historical data	Still need a query/analysis layer; may lag operational reality	Could be a *source* for SupplyNexus rather than a competitor
APIs	System-to-system access	Developer-only	Governed capability layer for users
Existing analytics / "control tower" style platforms	End-to-end visibility, alerts	Cost, implementation effort; depth of explanation **[UNKNOWN]**	**COMPETITIVE VALIDATION REQUIRED**
Search / internal knowledge bases	Document search	Keyword search; no synthesis; no link to live data	Grounded answers with citations
Traditional automation / rules / alerts	Deterministic detection of known conditions	Detects *that*, not *why*	Explanation of detected exceptions
Existing AI assistants (general-purpose assistants; AI features embedded in BI, ERP and productivity tools)	NL interaction; general knowledge; some NL-to-query	Generic domain understanding; governance varies; cross-system investigation depth **[UNKNOWN]**	Supply-chain specificity, evidence, governance — **COMPETITIVE VALIDATION REQUIRED**

12.1 Skeptical challenge — "embedded AI" threat

Problem	Vendors of ERP, WMS and BI platforms are adding natural-language and AI assistant features to their own products [GENERAL ENGINEERING KNOWLEDGE]. Simple NL querying may become a commodity feature.
Impact	A Data Assistant alone may not justify a separate product.
Example	A customer whose BI tool offers NL querying asks, "why do I need SupplyNexus to count returns?"
Recommendation	Differentiate on **cross-system investigation with evidence** and **supply-chain-specific knowledge**, not on NL querying alone. Validate against real alternatives before Phase 1 is locked. [ARCHITECTURAL RECOMMENDATION]
Alternative	Accept commodity NL querying as table-stakes and compete on domain depth and governance.
Trade-off	Investigation is harder and slower to build, but is where the defensible value likely lies.

13. AI Opportunities

Capability	1. Problem solved	2. Why AI is useful	3. Could deterministic software solve it?	4. Risk of using AI	5. Evidence required	6. MVP?
Knowledge retrieval (tenant SOPs, policies, definitions)	Finding and understanding organization-specific knowledge	Language understanding; synthesis across documents	Partially (keyword search finds documents; does not answer)	Hallucination; outdated documents; indirect prompt injection from documents	Cited passages / document references	**Yes — narrow corpus**
Generic supply-chain knowledge ("What is safety stock?")	Education / definitions	Fluent explanations	N/A — but a general-purpose assistant already does this	Low differentiation; answer may contradict tenant's own definition	Source documents if tenant-specific	**Only as part of tenant knowledge** (see 13.1)

Capability	1. Problem solved	2. Why AI is useful	3. Could deterministic software solve it?	4. Risk of using AI	5. Evidence required	6. MVP?
Natural-language querying	Non-SQL users need ad-hoc answers	Translates intent into a governed query	Partly: parameterised reports and filters cover many questions	Wrong filter/time range → confidently wrong number	The executed query/parameters and result rows	**Yes — for bounded question families**
Data analysis (ranking, trends, comparisons)	Summarising data	Explaining results in language	**Yes** — calculations must be deterministic	AI doing arithmetic incorrectly	Deterministic computation + data	AI **only** for intent and narration
Root-cause investigation	"Why did X happen?"	Hypothesis generation; multi-step evidence gathering; correlation	Partly: fixed diagnostic playbooks for known exception types	Plausible but wrong root cause; runaway cost; over-reach	Timeline, records and checks supporting each claim	**Yes — one bounded exception type**
Reasoning / correlation	Connecting evidence across systems	Flexible reasoning over heterogeneous evidence	Partially, for known patterns	Spurious correlation	Explicit evidence links	Within investigation only
Recommendations	"What should I do?"	Suggests next steps	Rules for known cases	Accountability; wrong advice	Evidence + policy references	**No (future)** — MVP may suggest *next investigation steps* only
Summarization	Condensing evidence and findings	Language generation	No	Omission of important facts	Link back to full evidence	Yes, as part of answers
Exception investigation	Triage of exceptions	Reasoning over varied cases	Rules detect exceptions	Same as investigation	Same	One type in MVP
Cross-system investigation	Joining evidence across systems	Reasoning over heterogeneous sources	Integration is deterministic; reasoning is not	Identifier mismatch → wrong conclusions	Source attribution per evidence item	Limited in MVP (single synthetic dataset)
Natural-language interface	Lower skill barrier	Language	Forms/filters can serve many needs	Ambiguity; misinterpretation	Show interpretation back to user	Yes

13.1 Skeptical challenge — "What is safety stock?"

Problem	The source's canonical knowledge example is a *generic* definition that any general-purpose AI assistant or textbook answers.
Impact	A Knowledge Assistant built around generic definitions has little product value.
Example	A user already gets a good definition of safety stock from a public assistant for free.
Recommendation	Re-frame the Knowledge capability as **"What is *our* safety-stock policy for category X, and where is it written?"** — tenant-specific, cited knowledge. [ARCHITECTURAL RECOMMENDATION]
Alternative	Keep generic knowledge as an onboarding / education feature.
Trade-off	Tenant-specific knowledge requires tenants to have useful documents (TO VALIDATE) and raises document-security concerns; generic knowledge is easy but weakly differentiated.

14. Non-AI Opportunities

Principle: Use AI where reasoning, language or ambiguity creates value. Use deterministic software where deterministic software is more reliable. [SOURCE-DERIVED FACT — "AI must not replace deterministic software where deterministic logic is sufficient."]

Area	Why deterministic	Status
KPI and metric calculations (return counts, backlog, fill rate, on-time %)	Must be exact, reproducible and consistent with BI	SOURCE-DERIVED FACT (principle) / INFERENCE (application)
Filtering, sorting, ranking ("highest backlog")	Pure data operations	INFERENCE
Authentication	Security-critical	SOURCE-DERIVED FACT
Authorization / permission checks / tenant isolation	Must never depend on model behaviour	SOURCE-DERIVED FACT
Data validation (inputs, identifiers like SKU-10023, ORD-458921)	Exact-match rules	SOURCE-DERIVED FACT (exact identifiers noted in source)
Business-rule enforcement (allocation rules, safety-stock formulas)	Deterministic business logic	INFERENCE
Transaction processing / write operations	Out of scope for MVP; must be explicit and approved	SOURCE-DERIVED FACT ("start with read-only")
Query safety (allowlists, read-only, no arbitrary queries)	Deterministic guardrails	SOURCE-DERIVED FACT
Routing of high-confidence requests	Rules first, model only when uncertain	SOURCE-DERIVED FACT
Exception detection for known conditions (e.g. orders > available stock)	Rule-based detection is exact	INFERENCE
Audit logging, rate limiting	Governance controls	SOURCE-DERIVED FACT

Implication for the product: In SupplyNexus, **AI interprets and explains; deterministic software computes, validates, authorizes and enforces.** [ARCHITECTURAL RECOMMENDATION]

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15. Core Use Cases

15.1 Minimum use cases

Field	UC-1 Supply Chain Knowledge	UC-2 NL Data Query	UC-3 Inventory Investigation (Oversell)	UC-4 Order Investigation	UC-5 Warehouse Investigation
Problem	Policies/SOPs/definitions hard to find and interpret	Ad-hoc data questions need SQL/BI skills	Explaining oversells/discrepancies is slow and manual	Explaining why a specific order was late/cancelled/split	Explaining backlog and throughput issues
User	Analyst, managers, new staff	Analyst, managers	Analyst, Inventory Mgr	Analyst, Customer Service	Warehouse Mgr, Ops Mgr
Trigger	"What is our policy for…?"	"How many orders were returned last month?"	"Why was SKU-10023 oversold?"	"Why is ORD-458921 late?"	"Which warehouse has the highest backlog and why?"
Current workflow	Search drives / ask colleagues	Dashboards / request to analyst / SQL	Section 11.1	Look up order in OMS, WMS, carrier	WMS dashboards + shift reports

Field	UC-1 Supply Chain Knowledge	UC-2 NL Data Query	UC-3 Inventory Investigation (Oversell)	UC-4 Order Investigation	UC-5 Warehouse Investigation
Desired outcome	Cited, tenant-specific answer	Correct number with visible query interpretation	Explanation + evidence timeline + confidence	Order timeline + cause	Ranking (deterministic) + driver explanation
Potential value	Time saved; consistent policy application	Self-service; analyst time saved	Faster root cause; fewer repeat incidents	Faster customer responses	Faster operational response
AI suitability	High	Medium (translation only)	High	Medium–High	Low (ranking) / Medium (why)
Data requirements	Tenant documents	Orders, returns, inventory, warehouse data	Order events, stock levels/movements over time, channel allocations, policies	Order, fulfilment, shipment events	Warehouse tasks, order backlog, capacity
Evidence requirements	Document citations	Query interpretation + result rows	Time-ordered stock vs order evidence; rule/policy references	Event timeline	Metrics + contributing factors
Complexity	Medium	Medium	High	Medium–High	Medium
MVP suitability	**Yes (narrow)**	**Yes (bounded families)**	**Yes — flagship investigation**	Candidate (second investigation)	Ranking yes (via UC-2); *why* later
Validation status	TO VALIDATE	TO VALIDATE	TO VALIDATE	TO VALIDATE	TO VALIDATE

15.2 Additional use cases evaluated

Use case	Summary	MVP suitability	Reason
Shipment Investigation	Why shipment SHP-98231 is delayed	Later	Requires carrier/TMS data not in initial datasets **[INFERENCE]**
Supplier Investigation	Why supplier performance dropped	Later	Requires procurement/ERP data
Returns Investigation	Why returns spiked for a category	Candidate (post-MVP)	Return-reason text analysis is AI-suitable; data availability TO VALIDATE
Procurement Investigation	Why PO-100234 is late; ASN mismatch	Later	ERP/EDI (e.g. EDI 856 / ASN) complexity
Transportation Investigation	Carrier performance, lane delays	Later	TMS data
Risk Investigation	Which SKUs/suppliers are at risk	Future	Borders on prediction/anomaly detection

16. MVP Discovery

16.1 Evaluation of the starting hypothesis

> **STARTING HYPOTHESIS** [SOURCE-DERIVED FACT]: (A) Knowledge Assistant → RAG; (B) Data Assistant → governed data/SQL tools; (C) Investigation Assistant → governed tools / MCP / agentic investigation. The source also states: "Do not implement all three immediately."

Note: the technology arrows (RAG, SQL, MCP, agents) are **candidate implementations**; product capabilities are evaluated here independently of them.

A. Knowledge Assistant

Dimension	Assessment
User value	Moderate if tenant-specific; low if generic
Business value	Moderate (consistency, onboarding, time to find policy)
Complexity	Medium
AI requirement	Genuine (language + retrieval)
Data requirement	Tenant documents (availability TO VALIDATE)
Risk	Hallucination, stale documents, document-borne prompt injection
Demonstrability	High
Learning value	High (retrieval, grounding, evaluation)
Strong justification	Real AI need; needed anyway to give investigations policy context
Weak justification	Generic definitions are commodity; alone it is "chat with documents"
Unknowns	Do target organizations have usable, current SOP/policy documents?
Dependencies	Document corpus (synthetic/public for MVP)
Validation	Ask practitioners how often they look up policies and how

B. Data Assistant

Dimension	Assessment
User value	High for non-SQL users; low for users with good dashboards
Business value	Moderate–High (analyst capacity)
Complexity	Medium–High (correctness, governance)
AI requirement	Partial — AI translates intent; computation is deterministic
Data requirement	Structured operational data
Risk	Confidently wrong numbers; commoditisation by embedded BI AI
Demonstrability	High
Learning value	High (governed data access, validation)
Strong justification	Foundation for investigation (investigations need governed data access); frequent need
Weak justification	Overlaps heavily with BI; alone is weakly differentiated
Unknowns	Volume of ad-hoc questions; tolerance for errors
Dependencies	Data model and dataset
Validation	Measure time-to-answer for ad-hoc questions today

C. Investigation Assistant

Dimension	Assessment
User value	Potentially highest (addresses the slowest workflow)
Business value	Potentially highest (faster exception resolution)
Complexity	High
AI requirement	Strongest genuine need (multi-step reasoning)
Data requirement	Time-series / event data across domains (not fully present in Olist — INFERENCE)
Risk	Wrong root cause, cost, latency, unbounded scope, trust
Demonstrability	Very high
Learning value	Very high
Strong justification	Core differentiation hypothesis; where fixed reports fail

Dimension	Assessment
Weak justification	"Investigate anything" is unbounded and unevaluable
Unknowns	Which exception type is most painful; what evidence users need to trust a root cause
Dependencies	**Depends on B** (governed data access) and benefits from **A** (policy context)
Validation	Walk through 3–5 real or realistic oversell cases with practitioners

16.2 Answers to the required discovery questions

Question	Answer	Status
Should all three capabilities exist?	Yes, in the **long-term product**.	INFERENCE
Should one be removed?	No — but **generic** knowledge should be removed from the MVP value proposition (keep tenant-specific knowledge).	ARCHITECTURAL RECOMMENDATION
Should one be delayed?	**Open-ended** investigation should be delayed. A **single bounded** investigation type should stay in MVP, otherwise the MVP collapses into "chat with BI + chat with docs," which is weakly differentiated.	ARCHITECTURAL RECOMMENDATION
Should the order change?	**Product-value order**: Data → Investigation → Knowledge. **Dependency/learning build order** (current SDLC: RAG → Data/SQL → MCP → Agents) is acceptable because Investigation depends on the other two. These are different orders and both are valid.	ARCHITECTURAL RECOMMENDATION
Should the MVP be smaller?	**Yes.** One beachhead, one primary persona, bounded question families, one investigation type, one dataset.	ARCHITECTURAL RECOMMENDATION
Is another capability more valuable?	Possibly **deterministic exception detection** ("show me today's oversells") as the entry point into investigation — it tells users *what* to investigate. It is non-AI and cheap.	ARCHITECTURAL RECOMMENDATION — TO VALIDATE
Is the problem sufficiently painful?	**Unknown.**	TO VALIDATE
Is AI actually required?	For knowledge synthesis and investigation reasoning — yes. For counts, rankings, KPIs, security — **no**.	INFERENCE
Where should deterministic software be preferred?	See Section 14.	SOURCE-DERIVED FACT + INFERENCE
What must be validated with real users?	Primary user, beachhead, most painful exception type, evidence needs, trust threshold, buyer.	TO VALIDATE

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17. Initial MVP Boundary

Proposed boundary [ARCHITECTURAL RECOMMENDATION — pending owner decision and validation]:

> **The SupplyNexus AI MVP helps a supply-chain/operations analyst at a single beachhead organization type (hypothesis: omnichannel retailer or distributor operating its own warehouses) to (1) get correct, traceable answers to a bounded set of operational data questions, (2) get cited answers from their organization's own supply-chain policies and procedures, and (3) investigate one exception type — inventory overselling — with an evidence-backed explanation. All access is read-only, tenant-scoped and governed. A human always decides what to do.**

MVP element	Included	Explicit limit
Governed Data Q&A	Orders, returns, inventory positions, warehouse backlog	Bounded question families; deterministic computation; interpretation shown to user

MVP element	Included	Explicit limit
Tenant Knowledge Q&A	Curated corpus of policies/SOPs/glossary (synthetic/public for MVP)	Cited answers only; "I don't know" when not grounded
Investigation	**One** type: inventory overselling (e.g. "Why was SKU-10023 oversold?")	Read-only; evidence-backed; no recommendations beyond suggested next checks
Exception list (deterministic)	Optional candidate: list of detected oversells to investigate	TO VALIDATE whether needed in MVP
Trust features	Evidence and source shown with every answer	—
Tenancy & security	Multi-tenant from the start; tenant-scoped everything	SOURCE-DERIVED requirement
Data	Public / synthetic / safe datasets only	No confidential enterprise data [SOURCE-DERIVED FACT]

Why this boundary:

1. It preserves the **differentiation hypothesis** (evidence-backed investigation) while keeping scope evaluable.
2. It respects the **dependency chain** (investigation needs governed data + policy context).
3. It is **buildable by an individual contributor** [SOURCE-DERIVED constraint].
4. Every element is **measurable** (correctness, groundedness, investigation success).
5. Oversell is the source's own flagship example and is well-suited to deterministic, seeded synthetic data. [INFERENCE]

Data caveat. Olist (a candidate dataset in the source) is e-commerce order data; it does not, to general knowledge, include warehouse stock-movement history needed to explain overselling. The MVP will likely require **deterministic synthetic inventory and warehouse data with seeded, known root causes** — which also makes investigations *evaluable* against known answers. [INFERENCE — TO VALIDATE in Phase 4]

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18. Product Scope

> This is **product** scope (what the product does for users), not implementation scope (what gets built in which phase).

18.1 In scope — current product hypothesis

Item	Status
Natural-language questions about supply-chain operations	SOURCE-DERIVED FACT
Distinguishing knowledge questions, data questions and investigations	SOURCE-DERIVED FACT
Governed, read-only access to supply-chain data	SOURCE-DERIVED FACT
Tenant-specific knowledge retrieval with citations	ARCHITECTURAL RECOMMENDATION (re-framing of source's Knowledge Assistant)
Evidence-backed investigation of a bounded exception type	SOURCE-DERIVED FACT (investigation) + ARCHITECTURAL RECOMMENDATION (bounded)
Evidence shown with answers	SOURCE-DERIVED FACT
Multi-tenant SaaS with tenant isolation	SOURCE-DERIVED FACT
Measurable AI quality (accuracy, groundedness, investigation success)	SOURCE-DERIVED FACT
Public / synthetic / safe data for initial product	SOURCE-DERIVED FACT
Domains: orders, returns, inventory, warehouse	INFERENCE (from source examples)

18.2 Out of scope — current product hypothesis

Item	Reason	Status
Write operations / actions in customer systems	Read-only first; writes need explicit approval & audit	SOURCE-DERIVED FACT
Shipment, supplier, procurement, transportation domains	Data availability; focus	ARCHITECTURAL RECOMMENDATION
Forecasting, predictive insights, what-if analysis	Future vision; requires foundational intelligence first	ARCHITECTURAL RECOMMENDATION
Open-ended "investigate anything"	Unbounded, unevaluable	ARCHITECTURAL RECOMMENDATION
Connectors to real enterprise systems (WMS/ERP/OMS/TMS)	Long-term vision; MVP uses safe datasets	SOURCE-DERIVED FACT (vision) + INFERENCE (MVP)
Billing, quotas, self-service onboarding	Phase 23 (SaaS Readiness)	SOURCE-DERIVED FACT (SDLC)
Confidential enterprise data	Explicitly prohibited	SOURCE-DERIVED FACT

19. Non-Goals

#	Non-goal	Basis
NG-1	**Not** an ERP, WMS, OMS or TMS replacement; SupplyNexus sits alongside systems of record	INFERENCE from "connect to" systems (source)
NG-2	**No** autonomous supply-chain execution	SOURCE-DERIVED (read-only first; writes require approval)
NG-3	**No** unrestricted database access or arbitrary queries by AI	SOURCE-DERIVED FACT
NG-4	**No** fully autonomous business decisions — humans remain accountable	ARCHITECTURAL / PRODUCT RECOMMENDATION (consistent with source human-in-the-loop direction)
NG-5	**Not** a generic consumer chatbot or general-purpose assistant	SOURCE-DERIVED ("not simply an AI demo"; "not a chatbot" — INFERENCE)
NG-6	**Not** every supply-chain domain at once	SOURCE-DERIVED ("do not build every possible supply-chain domain")
NG-7	**No** predictive forecasting before foundational intelligence exists	ARCHITECTURAL / PRODUCT RECOMMENDATION
NG-8	**Not** a showcase of every AI technique; no technology added only because it is popular	SOURCE-DERIVED FACT
NG-9	**Not** a replacement for BI dashboards for known, recurring KPIs	ARCHITECTURAL / PRODUCT RECOMMENDATION
NG-10	**No** AI-performed arithmetic for business metrics	ARCHITECTURAL / PRODUCT RECOMMENDATION (consistent with source deterministic principle)
NG-11	**No** cross-tenant data sharing or cross-tenant answers (including cached answers)	SOURCE-DERIVED FACT

20. Product Differentiation Hypotheses

> None of these is claimed as unique. All require **COMPETITIVE VALIDATION**.

#	Differentiation hypothesis	Why it may matter	Potential competitor overlap	What must be validated
D1	**Investigation, not just chat** — answers *why*, not only *what*	Fixed reports and dashboards rarely explain root causes	Control-tower / analytics platforms; AI features in BI	Do users value explanations enough to adopt a new tool?

#	Differentiation hypothesis	Why it may matter	Potential competitor overlap	What must be validated
D2	**Evidence-backed answers**	Operational decisions need trust; evidence enables verification	Many AI tools show citations for documents; fewer for data investigations **[UNKNOWN]**	What evidence format makes users trust an answer?
D3	**Governed access** (read-only, allowlisted, tenant-scoped, audited)	Security owners must approve AI access to operational data	Enterprise AI platforms emphasise governance	Is governance a buying criterion or a minimum requirement?
D4	**Supply-chain-specific intelligence** (domain terms, identifiers such as SKU/ORD/SHP/WH/PO/ASN/EDI 856)	Generic tools misunderstand domain language and identifiers	Vertical supply-chain platforms	Does domain specificity measurably improve answers?
D5	**Knowledge + data + investigation in one experience**	Real questions mix policy and data	BI + separate document search	Do users actually ask mixed questions?
D6	**Right tool for the question** (deterministic where possible, AI where needed)	Correctness and cost	Emerging in advanced AI products	Is this visible/valuable to users, or only an internal quality property?
D7	**Cross-system investigation**	The hardest part of today's workflow is crossing systems	Data platforms, integration vendors	Integration effort in real customers
D8	**Explainable findings with confidence**	Accountability and auditability	— **[UNKNOWN]**	Would users act on a "medium-confidence" finding?
D9	**Tenant-aware enterprise AI**	SaaS trust	Most enterprise SaaS	Likely table-stakes rather than a differentiator (INFERENCE)

Skeptical note. D3 and D9 are more likely **requirements to enter** the market than reasons to buy. The strongest differentiation candidates are **D1 + D2 + D4** combined. [INFERENCE]

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21. Business Value Hypotheses

> **No numerical ROI is claimed.** All magnitudes are **UNKNOWN / TO VALIDATE**.

#	Value hypothesis	Who benefits	How it could be measured	Status
V1	Reduced time to investigate an exception (e.g. an oversell)	Analyst, Ops Mgr	Time-on-task, before vs after	UNKNOWN / TO VALIDATE
V2	Increased analyst productivity (fewer manual extracts)	Analyst, Buyer	Ad-hoc requests handled per week	UNKNOWN / TO VALIDATE
V3	Faster operational decisions	Managers	Time from question to decision	UNKNOWN / TO VALIDATE
V4	Reduced manual reporting	Analyst	Hours spent on ad-hoc reports	UNKNOWN / TO VALIDATE
V5	Faster root-cause analysis → fewer repeat incidents	Ops, Inventory	Repeat-incident rate	UNKNOWN / TO VALIDATE
V6	Reduced information-search effort for policies	All users	Time to find a policy answer	UNKNOWN / TO VALIDATE
V7	Better visibility across systems	Managers	User-reported confidence; survey	UNKNOWN / TO VALIDATE

#	Value hypothesis	Who benefits	How it could be measured	Status
V8	Improved exception handling consistency	Ops	Variance in explanations; audit	UNKNOWN / TO VALIDATE
V9	Reduced shadow-AI risk (users pasting data into unapproved tools)	Security owner	Policy compliance	ASSUMPTION / TO VALIDATE

22. Success Metrics

> Metrics are defined here; **no targets** are set. Any number below is labelled **PROPOSED TARGET** and is for discussion only.

Category	Metric	Definition (discovery-level)	Target
Business	Investigation time	Time from question to evidence-backed explanation	UNKNOWN — baseline first
Business	Decision time	Time from exception to decision	UNKNOWN
Business	Manual effort	Manual extracts/reconciliations per investigation	UNKNOWN
User	Adoption	Active users / invited users	UNKNOWN
User	Task completion	% of sessions where the user got a usable answer	UNKNOWN
User	Repeat usage	Users returning weekly	UNKNOWN
User	Trust	% of answers users mark as useful/correct; evidence opened	UNKNOWN
Product	Questions answered	By type (knowledge / data / investigation)	—
Product	Investigations completed	Investigations reaching an evidence-backed conclusion	—
Product	Evidence usage	% of answers where the user inspected evidence	—
Product	Correct refusal	% of out-of-scope/unanswerable questions correctly declined	—
AI	Groundedness / faithfulness	Knowledge answers supported by cited sources	PROPOSED TARGET: set after baseline on golden dataset
AI	Retrieval accuracy	Relevant document retrieved for knowledge questions	PROPOSED TARGET: set after baseline
AI	Data answer correctness	Answer matches known correct result on golden questions	PROPOSED TARGET: very high bar — wrong numbers are unacceptable
AI	Question-type routing correctness	Right capability chosen	PROPOSED TARGET: set after baseline
AI	Tool correctness	Right governed capability with right parameters	PROPOSED TARGET: set after baseline
AI	Investigation success	Correct root cause on seeded synthetic cases	PROPOSED TARGET: set after baseline
AI	Safety	Prompt-injection resistance; zero cross-tenant leakage	**Zero** cross-tenant leakage is a requirement, not a target (SOURCE-DERIVED FACT)
Operational	Latency	Time to answer by question type	UNKNOWN — set in Phase 1/5
Operational	Cost per request	Model + infrastructure cost per answer	UNKNOWN — set in Phase 1/5
Operational	Failure rate	Errors / requests	UNKNOWN

23. Assumptions Register

ID	Assumption	Why we believe it	Impact if false	Confidence	Validation method	Status
A-01	Supply-chain information is fragmented across multiple systems in target organizations	Source thesis; general industry knowledge	Core problem weakens	Medium	Practitioner interviews	ASSUMPTION
A-02	Ad-hoc and *why* questions are frequent and time-consuming	Source examples	Low usage; weak value	Low	Interviews; time-on-task	TO VALIDATE
A-03	The supply-chain/operations analyst is the primary user	Source canvas	Wrong UX and priorities	Low–Medium	Interviews	TO VALIDATE
A-04	Head of Supply Chain/Operations is the economic buyer	Inference	Wrong value messaging	Low	Buyer conversations	UNKNOWN
A-05	Users will trust AI answers if evidence is shown	Inference	Adoption fails	Low	Prototype testing	TO VALIDATE
A-06	Existing BI/dashboards do not adequately answer *why* questions	General knowledge	Differentiation weakens	Medium	Tool walkthroughs	TO VALIDATE
A-07	Overselling is a sufficiently painful and frequent exception	Source example	Wrong flagship investigation	Low	Interviews	TO VALIDATE
A-08	Target organizations have useful policy/SOP documents	Inference	Knowledge capability low value	Low	Interviews; document samples	TO VALIDATE
A-09	Security teams will approve governed AI access to operational data	Inference	Sales blocked	Low	Security-owner conversations	TO VALIDATE
A-10	Synthetic/public data can realistically demonstrate value	Source (public/synthetic data)	Demo not convincing	Medium	Prototype review with practitioners	TO VALIDATE
A-11	AI cost per request can be kept acceptable	Inference	Unviable economics	Unknown	Phase 5 cost modelling	UNKNOWN
A-12	A single individual can build a credible MVP within the bounded scope	Source constraint	MVP never ships	Medium (if bounded)	Phase 1 scoping	ASSUMPTION
A-13	Multi-tenancy is required from MVP	Source principle	Extra cost if only one pilot customer	**Known as a project decision**	—	KNOWN (source decision)
A-14	Read-only access is sufficient for MVP value	Source; inference	Users demand actions	Medium	Interviews	TO VALIDATE
A-15	Users prefer natural language over filters/forms for ad-hoc questions	Source vision	Simpler UI would suffice	Low	Prototype A/B	TO VALIDATE

24. Validation Strategy

24.1 Methods

Method	Validates	Feasibility for an individual contributor	Notes
Supply-chain practitioner / expert interviews	Problem, pain, users, workflow, evidence needs	Feasible (professional network)	Do **not** use confidential employer data
Analyst interviews	Workflow, time-on-task, tools	Feasible	Ask about last real investigation
Workflow observation (walkthrough)	Current workflow hypothesis	Partially feasible	Screen-share of a sanitised past case
Prototype testing (clickable or thin)	Trust, evidence format, NL preference	Feasible after Phase 9–12	—
Synthetic-data experiments	Investigation correctness on seeded root causes	Highly feasible	Also becomes evaluation data
Public datasets (e.g. Olist)	Data Q&A realism	Feasible	Limited inventory/warehouse depth
Pilot users	Adoption, value	Later (post-MVP)	Requires security approval
Time-on-task measurement	V1–V4	Feasible in prototype testing	Baseline first
Before/after workflow comparison	Business value	Pilot stage	—
Desk research on existing solutions	Differentiation	Feasible	Required before Phase 1 is locked

24.2 Questions to answer before Phase 1

1. Which organization type is the beachhead?
2. Who is the primary user, and who is the economic buyer?
3. Which exception/investigation type is most painful and frequent?
4. How long does that investigation take today, and what tools are used?
5. What evidence would make a user trust an AI explanation?
6. Do existing BI/ERP/WMS tools (including their AI features) already solve the target questions?
7. Which data domains are essential for the first investigation?
8. Is tenant-specific knowledge (policies/SOPs) actually available and valuable?
9. What level of AI autonomy is acceptable (answer only, recommend, act)?
10. What single measurable outcome matters most to the buyer?

24.3 Pragmatic path given the learning-platform context

Because SupplyNexus AI is built by an individual contributor and doubles as a learning platform [SOURCE-DERIVED FACT], full customer discovery may not be possible before Phase 1. Two legitimate options exist:

Option	Description	Trade-off
1. Light validation sprint (recommended)	3–5 conversations with supply-chain practitioners + desk research on existing solutions before Phase 1	Small delay; much stronger PRD
2. Declared-assumption decision	Owner explicitly accepts key assumptions (A-02, A-03, A-07) as working hypotheses, records this in ADR-0001, and proceeds	Faster; risk that the PRD optimises for an unvalidated problem

25. Major Risks

> Likelihood is described qualitatively; no numerical likelihoods are claimed.

#	Risk	Impact	Likelihood (qualitative)	Evidence	Mitigation hypothesis	Validation requirement
R1	Problem not painful enough	Low adoption	Unknown	Pain magnitude UNKNOWN in source	Focus on *why*/exception questions	Interviews, time-on-task
R2	Users do not trust AI answers	Low adoption	Plausible	General industry experience	Evidence with every answer; show interpretation; refuse when unsure	Prototype testing
R3	Inaccurate answers (wrong numbers, wrong root causes)	Loss of trust; wrong decisions	Plausible	General engineering knowledge	Deterministic computation; validation; golden datasets	AI evaluation
R4	Hallucination in knowledge answers	Wrong policy application	Plausible	General knowledge	Grounding + citations + "I don't know"	Groundedness evaluation
R5	Poor data quality in customer systems	Wrong conclusions	Plausible	General knowledge	Surface data-quality issues as evidence	Pilot
R6	Integration complexity	Slow onboarding; high cost	Likely for real customers	Many heterogeneous systems (source)	MVP on safe datasets; integrations later	Phase 7
R7	Security concerns block adoption	Sales blocked	Plausible	Security owner concerns	Governed, read-only, audited design	Security-owner conversations
R8	Cross-tenant data leakage	Catastrophic trust loss	Low if designed; catastrophic impact	Source: never allow	Tenant isolation everywhere (Phase 6)	Security testing
R9	Prompt injection / indirect injection via documents	Data exposure, wrong answers	Plausible	Source identifies it	Guardrails (Phase 6)	Adversarial evaluation
R10	AI cost too high	Unviable pricing	Unknown	—	Deterministic-first routing; bounded investigations	Cost measurement
R11	Latency too high for investigations	Poor UX	Plausible	Multi-step reasoning	Bounded steps; progressive results	Performance testing
R12	Overly broad MVP	MVP never ships	**High** without scoping	Long technology list; 3 capabilities; individual builder	Section 17 boundary	Phase 1 scope review
R13	Weak differentiation	No reason to buy	Plausible	Embedded AI in existing tools	D1+D2+D4 focus	Competitive research
R14	Users prefer existing BI tools	Low adoption	Plausible	—	Complement, not replace, BI	Interviews
R15	Governance/compliance requirements (AI governance, data residency)	Delays	Unknown	—	Auditability, evaluation evidence	Buyer conversations

#	Risk	Impact	Likelihood (qualitative)	Evidence	Mitigation hypothesis	Validation requirement
R16	Learning goals drive scope ("technology showcase" drift)	Complexity, no product focus	**Plausible**	Long list of AI concepts in source; source also warns against it	Problem → requirement → decision → ADR discipline	Phase reviews
R17	Synthetic data makes demos unconvincing	Weak validation	Plausible	—	Realistic seeded scenarios reviewed by practitioners	Practitioner review

26. Product Principles

#	Principle	Basis
PP-1	**Solve a real supply-chain problem before introducing AI.**	SOURCE-DERIVED
PP-2	**Important answers carry evidence.** Users can see where an answer came from.	SOURCE-DERIVED
PP-3	**AI augments people; humans own decisions.**	ARCHITECTURAL / PRODUCT RECOMMENDATION
PP-4	**Deterministic logic stays deterministic.** AI interprets and explains; software computes, validates and enforces.	SOURCE-DERIVED
PP-5	**Security and tenant isolation are foundational**, not features.	SOURCE-DERIVED
PP-6	**The MVP stays focused**: one beachhead, one primary persona, one investigation type.	SOURCE-DERIVED (focus) + RECOMMENDATION (specifics)
PP-7	**Every major capability has measurable value.**	SOURCE-DERIVED
PP-8	**Complexity must be justified** by a problem, never by popularity.	SOURCE-DERIVED
PP-9	**Enterprise trust is a product requirement.**	SOURCE-DERIVED
PP-10	**AI is governed, never unrestricted.** Read-only first; actions require approval.	SOURCE-DERIVED
PP-11	**Say "I don't know"** rather than guess.	ARCHITECTURAL / PRODUCT RECOMMENDATION
PP-12	**Complement existing systems**; do not replace systems of record or BI.	ARCHITECTURAL / PRODUCT RECOMMENDATION

27. Future Product Vision

> Everything in this section is **FUTURE VISION** and must **not** expand the MVP.

Horizon	Capability	Description
MVP	Governed data Q&A; tenant knowledge Q&A; oversell investigation	Section 17
Post-MVP 1	More investigations	Order, warehouse backlog *why*, returns spikes
Post-MVP 1	Real connectors	Governed connectors to WMS/ERP/OMS/TMS, databases, APIs, files, object storage
Post-MVP 2	Domain intelligence	Inventory, order, warehouse, supplier, shipment, returns, procurement intelligence
Post-MVP 2	Proactive detection	Anomaly and risk detection; exceptions surfaced before users ask

Horizon	Capability	Description
Post-MVP 3	Recommendations	Evidence-backed recommended actions
Post-MVP 3	Predictive & what-if	Forecasting, scenario analysis
Long-term	Controlled actions	Human-approved, audited write actions

Guardrail: a future capability enters scope only through the same discipline: *problem → requirement → alternatives → trade-offs → decision → implementation → measurement → ADR*. [SOURCE-DERIVED FACT]

28. 30-Second Product Explanation

> SupplyNexus AI is a supply-chain intelligence assistant for operations and supply-chain analysts. When something goes wrong — an item gets oversold, a warehouse backs up, orders run late — the answer is usually spread across several systems, documents and people, and finding it takes hours of manual work. SupplyNexus lets analysts ask in plain language and get answers backed by evidence from their own data and policies, including *why* an exception happened. It is governed and read-only by design, so organizations can trust what it accesses and what it says.

29. 2-Minute Product Explanation

> **The problem.** Supply-chain operations run on many systems — order management, warehouse management, ERP, transportation — plus policies and procedures in documents. Dashboards tell you *what* is happening, but when a manager asks *why* SKU-10023 was oversold or why a warehouse is backed up, an analyst has to export data from several systems, reconcile it in spreadsheets, check policies and chase other teams. That investigation is slow, manual and depends on a few experts. *(This is our working hypothesis; we are validating how painful it is.)*
>
> **The product.** SupplyNexus AI is a multi-tenant SaaS assistant that answers three kinds of questions differently. Factual data questions — "how many orders were returned last month?" — are answered by governed, read-only data access with deterministic calculations, so the numbers are exact. Knowledge questions — "what is *our* safety-stock policy?" — are answered from the organization's own documents, with citations. And investigation questions — "why was this SKU oversold?" — are handled by a bounded investigation that gathers evidence step by step and explains the likely cause, showing the evidence behind it.
>
> **Why it's different.** It is not a generic chatbot: it answers from the customer's governed data and documents, shows its evidence, uses AI only where reasoning or language is needed, and never has unrestricted access to data.
>
> **The MVP.** One primary user (the analyst), one organization type, bounded data questions, a curated knowledge base, and one investigation type — overselling — on public and synthetic data.

30. 5-Minute Product Explanation

1. Problem. Supply-chain teams face a steady stream of operational exceptions: overselling, stockouts, inventory discrepancies, warehouse backlog, shipment delays, returns spikes. The information needed to explain them is fragmented across order, warehouse, inventory, ERP and transportation systems, and across documents

and people. Existing dashboards are good at known KPIs but weak at ad-hoc and *why* questions. *(Hypothesis — magnitude to validate.)*

2. Users. Our primary user hypothesis is the supply-chain or operations analyst — the person who fields ad-hoc questions and investigates exceptions. The main consumers of their work are operations, inventory and warehouse managers. IT, data and security owners are essential stakeholders because they must approve access.

3. Today's workflow. Question arrives → decide which systems → open reports → export → combine in Excel or SQL → contact other teams → check policies → form hypotheses → prepare an explanation. Much of this is manual, repetitive and cross-system; the reasoning step is the only part that fundamentally needs human-like judgement.

4. Product. SupplyNexus AI is a natural-language assistant that first works out what kind of question it is facing. It does not treat everything as an AI problem.

5. Data value. For factual questions, AI only interprets the question; the actual answer comes from governed, read-only data access with deterministic calculations. The user sees how their question was interpreted so they can spot mistakes.

6. Knowledge value. For policy and procedure questions, answers come from the organization's own documents, with citations — not from general model knowledge.

7. Investigation. For *why* questions, SupplyNexus runs a bounded investigation: it gathers evidence through approved capabilities, correlates it, and explains the most likely cause with the evidence attached. In the MVP, this is limited to one exception type — overselling — so that we can measure whether it gets the right answer.

8. Evidence. Every important answer carries its evidence: records, calculations, document passages, timelines. Trust is treated as a product requirement.

9. Business value. Hypotheses: shorter investigation time, more analyst capacity, faster decisions, more consistent explanations. We have not measured any of these yet; we will baseline them.

10. Governance. Multi-tenant from day one, strict tenant isolation, read-only access, no arbitrary queries, auditing. These are what allow a security team to say yes.

11. MVP. One beachhead organization type, one primary persona, bounded data questions, a curated knowledge base, one investigation type, public and synthetic data.

12. Future vision. More investigation types, real system connectors, proactive anomaly detection, recommendations, predictive and what-if analysis, and eventually human-approved actions — each added only when a real problem justifies it.

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31. Phase 0 Decision Summary

PHASE 0 DECISION SUMMARY

Item	Current position	Status
Current product hypothesis	A governed, evidence-backed supply-chain intelligence assistant that answers data, knowledge and investigation questions differently	SOURCE-DERIVED + INFERENCE
Primary problem	Slow, manual, cross-system investigation of operational questions and exceptions due to fragmented data and knowledge	INFERENCE — pain magnitude TO VALIDATE
Primary user hypothesis	Supply-chain / operations analyst	TO VALIDATE
Primary consumer	Operations / supply-chain manager	TO VALIDATE

Item	Current position	Status
Target organization hypothesis	Beachhead: omnichannel retailer or distributor operating its own warehouses	TARGET MARKET HYPOTHESIS
Economic buyer	Head of Supply Chain / Operations (hypothesis)	UNKNOWN
Core use cases	UC-1 tenant knowledge; UC-2 governed data Q&A; UC-3 oversell investigation	RECOMMENDATION
MVP hypothesis	Section 17 boundary — reduced from the original three open-ended capabilities to bounded versions	RECOMMENDATION — owner decision required
Starting hypothesis verdict	Keep all three capabilities, **bounded**; reframe Knowledge as tenant-specific; limit Investigation to one type; Data is foundational	RECOMMENDATION
Non-goals	Section 19 (NG-1 … NG-11)	SOURCE-DERIVED + RECOMMENDATION
Differentiation hypotheses	Strongest: investigation (D1) + evidence (D2) + supply-chain specificity (D4)	HYPOTHESIS — COMPETITIVE VALIDATION REQUIRED
Business-value hypotheses	V1–V9; no ROI claimed	UNKNOWN / TO VALIDATE
Success metrics	Defined across business, user, product, AI, operational categories; no targets yet	DEFINED — targets UNKNOWN
Major assumptions	A-02 (frequent *why* questions), A-03 (analyst is primary user), A-05 (evidence builds trust), A-07 (oversell is painful)	TO VALIDATE
Validation requirements	Section 24.2 ten questions	OPEN
Major risks	R12 broad MVP, R16 technology-showcase drift, R1 insufficient pain, R13 weak differentiation, R3/R4 inaccuracy, R8 tenant leakage	IDENTIFIED
Open questions	Section 32	OPEN

Key product decisions proposed in this phase

ID	Decision	Type
PD-01	SupplyNexus AI is a **decision-support / investigation** product, not a system of execution	Proposed
PD-02	The three capabilities are retained, but **bounded** for the MVP	Proposed
PD-03	Knowledge capability is **tenant-specific and cited**, not generic definitions	Proposed
PD-04	MVP includes **one** investigation type: inventory overselling	Proposed
PD-05	AI interprets and explains; **deterministic software computes, authorizes and enforces**	Source-derived
PD-06	**Read-only**, governed, tenant-scoped access in MVP	Source-derived
PD-07	MVP uses **public / synthetic / safe data** only, likely including seeded synthetic inventory scenarios	Source-derived + Proposed
PD-08	Product-value priority (Data → Investigation → Knowledge) is distinct from build order (current SDLC order retained)	Proposed
PD-09	Single primary persona (analyst) and single beachhead organization type	Proposed — TO VALIDATE

These decisions should be recorded in **ADR-0001 — Product Discovery** once the owner accepts or amends them. [SOURCE-DERIVED: the source anticipates `docs/decisions/ADR-0001-product-discovery.md`] ADR-0001 is **not** created by this document.

32. Open Questions

OPEN QUESTIONS BEFORE PHASE 1

#	Question	Why it blocks Phase 1	Owner
OQ-01	Who is the first target customer (beachhead organization type)?	Determines use cases, data domains and language	Product owner
OQ-02	Who is the economic buyer?	Determines which outcome the PRD optimises	Product owner
OQ-03	Which workflow/exception is most painful? Is overselling the right flagship investigation?	Determines the MVP investigation	Validation
OQ-04	Which capability should the MVP *lead* with in positioning?	Affects PRD priorities	Product owner
OQ-05	Which data sources/domains are essential for the first investigation?	Drives data requirements (Phase 1 → Phase 4)	Product owner + validation
OQ-06	What evidence do users require to trust an answer?	Defines answer/evidence requirements	Validation
OQ-07	What level of AI autonomy is acceptable (answer / recommend / act)?	Defines boundaries & human-in-the-loop requirements	Validation
OQ-08	What single measurable business outcome matters most?	Defines primary success metric	Buyer validation
OQ-09	Is tenant-specific knowledge available and valuable in target organizations?	Determines Knowledge capability value	Validation
OQ-10	Do existing tools (including embedded AI features) already solve the target questions?	Determines differentiation	Desk research
OQ-11	Will a light validation sprint be done, or will assumptions be formally accepted (Section 24.3)?	Determines Phase 0 exit	Product owner
OQ-12	Accept, amend or reject the proposed MVP boundary (Section 17) and decisions PD-01 … PD-09?	Required input to PRD	Product owner

33. Phase 1 Inputs

PHASE 1 — REQUIREMENTS & PRD INPUTS

> Phase 1 is **not** performed here. This section only lists what Phase 1 must receive.

#	Input	Source section
1	Product vision and working thesis	§2, §3
2	Problem statement (with confidence and validation needs)	§4, §5
3	Beachhead organization hypothesis	§6
4	Primary persona and secondary personas	§7, §8
5	Buyer / stakeholder map, including security & IT concerns	§9
6	Prioritised pain points	§10
7	Current workflow hypothesis	§11
8	Existing-solution landscape (with competitive validation gap)	§12
9	AI vs non-AI boundary	§13, §14
10	Core use cases UC-1, UC-2, UC-3 (+ candidates)	§15

#	Input	Source section
11	Proposed MVP boundary and rationale	§16, §17
12	Product scope and non-goals	§18, §19
13	Differentiation and business-value hypotheses	§20, §21
14	Success metric definitions (targets to be set)	§22
15	Assumptions register	§23
16	Validation questions and chosen validation path	§24
17	Risk register	§25
18	Product principles	§26
19	Proposed product decisions PD-01 … PD-09 (owner-accepted version)	§31
20	Resolved or explicitly accepted open questions	§32

Phase 0 Final Quality Check

Check	Result
Product problem is clear	✓ (magnitude TO VALIDATE)
Target user clear or marked TO VALIDATE	✓ TO VALIDATE
Target organization clear or marked TO VALIDATE	✓ TARGET MARKET HYPOTHESIS
Buyer identified or marked UNKNOWN	✓ UNKNOWN (hypothesis stated)
Business value defined as hypotheses	✓
Core use cases identified	✓
AI opportunities justified	✓
Non-AI opportunities identified	✓
MVP bounded	✓ (proposed)
Non-goals explicit	✓
Differentiation treated as hypothesis	✓
Success metrics exist	✓ (no targets)
Assumptions documented	✓
Validation questions exist	✓
Risks documented	✓
Future vision separated from MVP	✓
30-second / 2-minute / 5-minute explanations	✓
Phase 1 inputs identified	✓
No application code, schema, API or deployment design	✓
No unsupported customer claims presented as facts	✓

Phase 0 Status

PHASE 0 STATUS: CONTINUE PRODUCT DISCOVERY

Why not "Ready for Phase 1": the document is complete as a *discovery artifact*, but the source's own completion rule says Phase 0 must not be declared complete simply because documents exist. Three items that the PRD depends on remain unresolved:

1. **Buyer** is UNKNOWN (OQ-02).
2. **Pain magnitude** and the **flagship investigation** (overselling) are unvalidated (OQ-03).

3. The **proposed MVP boundary and product decisions** have not yet been accepted by the product owner (OQ-12).

What moves Phase 0 to READY FOR PHASE 1:

- The owner accepts/amends the MVP boundary and PD-01 … PD-09; **and**
- Either (a) a light validation sprint (3–5 practitioner conversations + desk research) is completed, **or** (b) the owner formally accepts the key assumptions (A-02, A-03, A-05, A-07) as working hypotheses and records this in ADR-0001.

Either route is legitimate for a learning-platform project; route (b) must be explicit so that the PRD is honest about what is unvalidated.

SUPPLYNEXUS AI — PHASE 0 HANDOFF

CURRENT STATE

Phase 0 discovery document v0.1 drafted. No code, repository structure, architecture or technology decisions exist. Status: **CONTINUE PRODUCT DISCOVERY** pending owner decisions (OQ-11, OQ-12) and buyer/pain validation.

WHAT WE DISCOVERED

- The source thesis (fragmented data and knowledge make investigation hard) is plausible but unvalidated.
- The strongest AI value lies in **investigation** and **tenant-specific knowledge**; counts, rankings and KPIs should be deterministic.
- Generic knowledge ("What is safety stock?") is weakly differentiated; tenant-specific, cited knowledge is the useful version.
- NL data querying alone risks commoditisation by AI features in existing tools.
- The original three-capability MVP is directionally right but **too broad** unless each capability is bounded.
- Investigation depends on governed data access; product-value order and build order are different and both valid.
- Public data (Olist) likely lacks the inventory depth needed for oversell investigation; seeded synthetic data is likely required and doubles as evaluation data.

PRODUCT VISION

A governed, evidence-backed supply-chain intelligence assistant that helps teams understand, query and investigate their operations — evolving from knowledge and data answers, to investigations, to recommendations and, eventually, approved actions.

PROBLEM STATEMENT

Supply-chain and operations teams cannot quickly obtain trustworthy answers and explanations about their operation because data and knowledge are spread across systems, documents and people; explaining *why* an exception occurred is especially slow and manual. (Magnitude TO VALIDATE.)

TARGET USERS

Primary: supply-chain / operations analyst (TO VALIDATE). Secondary: operations/supply-chain, inventory and warehouse managers. Occasional: logistics, procurement, executives. Technical/administrative: data analysts, IT/data team, tenant admins.

TARGET ORGANIZATIONS

TARGET MARKET HYPOTHESIS — beachhead: omnichannel retailer or distributor operating its own warehouses. Others (3PLs, manufacturers, logistics, warehouse operators) are later candidates.

PERSONAS

Priya (Analyst — primary), Daniel (Ops/SC Manager — primary consumer), Marcus (Inventory Manager), Elena (Warehouse Manager), Sam (IT/Data Owner), Executive/Director. All hypothetical.

BUYERS / STAKEHOLDERS

Economic buyer UNKNOWN (hypothesis: Head of Supply Chain/Operations). Critical gatekeepers: security owner, IT owner, data owner, AI governance owner.

CORE USE CASES

UC-1 Tenant supply-chain knowledge (cited) · UC-2 Governed NL data query (bounded families) · UC-3 Inventory oversell investigation (flagship). Candidates: order, warehouse *why*, returns investigation.

AI VS NON-AI USE CASES

AI: knowledge synthesis, NL intent interpretation, investigation reasoning/correlation, summarisation. Non-AI: calculations, filtering/ranking, authentication, authorization, tenant isolation, validation, business rules, known-exception detection, query safety, audit.

MVP

Analyst at one beachhead; governed read-only data Q&A for orders/returns/inventory/warehouse backlog; cited tenant knowledge Q&A over a curated corpus; one investigation type (overselling) with evidence; multi-tenant; public/synthetic data only.

NON-GOALS

No ERP/WMS/OMS/TMS replacement; no autonomous execution; no unrestricted data access; no autonomous decisions; no generic chatbot; not every domain; no forecasting yet; no technology showcase; no BI replacement; no AI arithmetic; no cross-tenant sharing.

PRODUCT DIFFERENTIATION HYPOTHESES

Strongest: investigation (D1) + evidence (D2) + supply-chain specificity (D4). Governance and tenancy are likely table-stakes. COMPETITIVE VALIDATION REQUIRED.

BUSINESS VALUE

Hypotheses V1–V9 (investigation time, analyst productivity, decision speed, reduced manual reporting, faster root cause, reduced search effort, visibility, consistency, reduced shadow-AI). No ROI claimed.

SUCCESS METRICS

Business, user, product, AI and operational metrics defined; targets to be set after baselining. Zero cross-tenant leakage is a requirement.

ASSUMPTIONS

A-01 … A-15; most critical: A-02, A-03, A-05, A-07, A-04 (buyer).

VALIDATION ITEMS

Section 24.2 (ten questions); choose validation sprint or declared-assumption route.

RISKS

Most significant: broad MVP (R12), technology-showcase drift (R16), insufficient pain (R1), weak differentiation (R13), inaccurate answers (R3/R4), tenant leakage (R8), prompt injection (R9), cost/latency (R10/R11).

PRODUCT PRINCIPLES

PP-1 … PP-12 (problem before AI; evidence; humans decide; deterministic stays deterministic; security & tenancy foundational; focused MVP; measurable value; justified complexity; trust; governed AI; say "I don't know"; complement existing systems).

KEY PRODUCT DECISIONS

PD-01 … PD-09 (proposed; PD-05, PD-06 are source-derived). To be confirmed by the owner and recorded in ADR-0001.

OPEN QUESTIONS

OQ-01 … OQ-12, most critically OQ-02 (buyer), OQ-03 (flagship pain), OQ-11 (validation route), OQ-12 (accept MVP boundary).

PHASE 1 INPUTS

Section 33 — twenty inputs, including the owner-accepted MVP boundary and decisions.

NEXT OBJECTIVE

Close OQ-11 and OQ-12; complete light validation or record accepted assumptions in ADR-0001; then transition Phase 0 to READY FOR PHASE 1.

NEXT PHASE: PHASE 1 — REQUIREMENTS & PRD (not started)